3. The number of ways selecting 4 players out of 5 is

(a) 4! (b) 20 (c) 25 (d) 5

4. If $nP_r = 720(nC_r)$, then r is equal to

(a) 4 (b) 5 (c) 6 (d) 7

5. The possible out comes when a coin is tossed five times

(a) 2^5 (b) 5^2 (c) 10 (d) $\frac{5}{2}$

6. The number of diagonals in a polygon of n sides is equal to

(a) nC_2 (b) $nC_2 - 2$

(c) $nC_2 - n$ (d) $nC_2 - 1$

7. The greatest positive integer which divide $n(n+1)(n+2)(n+3)$ for all $n \in N$ is

(a) 2 (b) 6 (c) 20 (d) 24

8. If n is a positive integer, then the number of terms in the expansion of $(x+a)^n$ is

(a) n (b) $n+1$

(c) $n-1$ (d) $2n$

9. For all $n > 0$, $nC_1 + nC_2 + nC_3 + \ldots + nC_n$ is equal to

(a) 2^n (b) $2^n - 1$

(c) n^2 (d) $n^2 - 1$

10. The term containing x^3 in the expansion of $(x-2y)^7$ is

(a) 3rd (b) 4th (c) 5th (d) 6th

11. The middle term in the expansion of $\left(x+\frac{1}{x}\right)^{10}$ is

(a) $10C_4\left(\frac{1}{x}\right)$ (b) $10C_5$

(c) $10C_6$ (d) $10C_7 x^4$

12. The constant term in the expansion of $\left(x+\frac{2}{x}\right)^6$ is

(a) 156 (b) 165

(c) 162 (d) 160

13. The last term in the expansion of $(3+\sqrt{2})^8$ is

(a) 81 (b) 16

(c) $8\sqrt{2}$ (d) $27\sqrt{3}$

14. If $\frac{kx}{(x+4)(2x-1)} = \frac{4}{x+4} + \frac{1}{2x-1}$ then k is equal to

(a) 9 (b) 11 (c) 5 (d) 7

15. The number of 3 letter words that can be formed from the letters of the word number when the repetition is allowed are

(a) 206 (b) 133 (c) 216 (d) 300

16. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines is

(a) 18 (b) 12 (c) 9 (d) 6

17. There are 10 true or false questions in an examination. Then these questions can be answered in

(a) 240 ways (b) 120 ways

(c) 1024 ways (d) 100 ways

18. The value of $(5C_0 + 5C_1) + (5C_1 + 5C_2) + (5C_2 + 5C_3) + (5C_3 + 5C_4) + (5C_4 + 5C_5)$ is

(a) $2^6 - 2$ (b) $2^5 - 1$

(c) 2^8 (d) 2^7

19. The total number of 9 digit number which have all different digit is

(a) 10! (b) 9!

(c) $9 \times 9!$ (d) $10 \times 10!$

02_11th_BM-STAT_Ch-2-EM.indd 46 21-04-2020 12:19:09 PM